

Vertical dilations

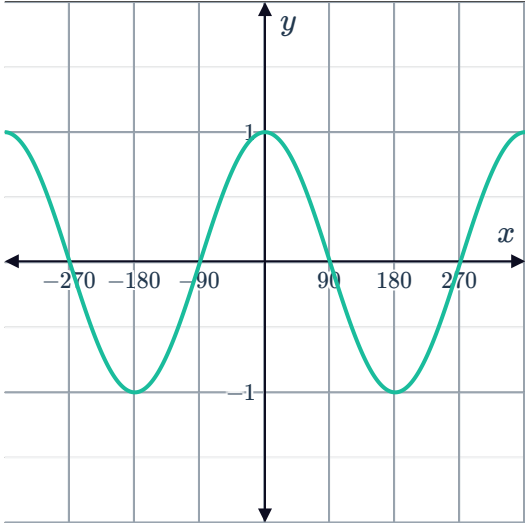


1

a

θ	0	60	90	120	180	240	270	300	360
$\cos \theta$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

b



c 1

d -1

e $[-4, 4]$

2

a $(-\infty, \infty)$

b $[-1, 1]$

3

a $(-\infty, \infty)$

b $[-2, 2]$

4 8

5

a $y = 2 \sin x$

b $y = -\sin x$

c $y = 3 \cos x$

d $y = -1.5 \cos x$

e $y = 3 \sin x$

f $y = 2.5 \sin x$

6 270

7

a 8

b 0

c -8

8 5

9 $-2 \sin x$

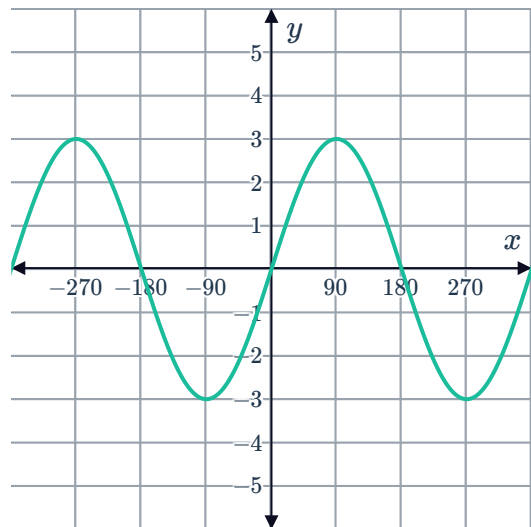


10

a

i $A = 3$

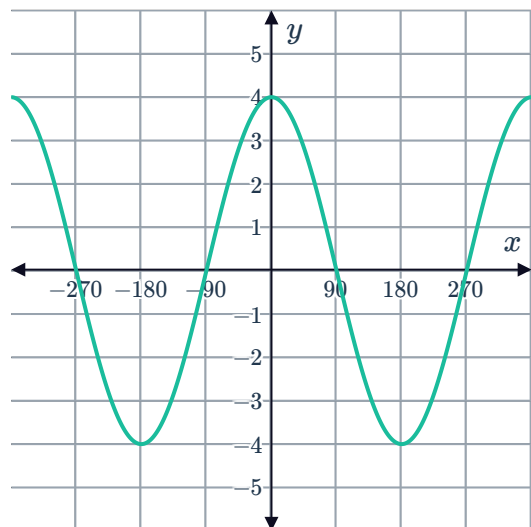
ii



b

i $A = 4$

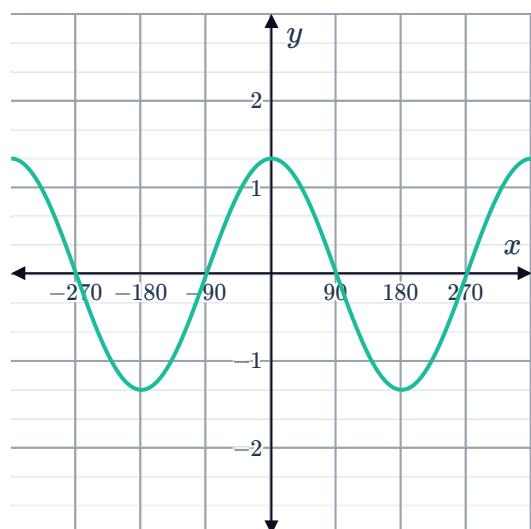
ii



c

i $A = \frac{4}{3}$

ii

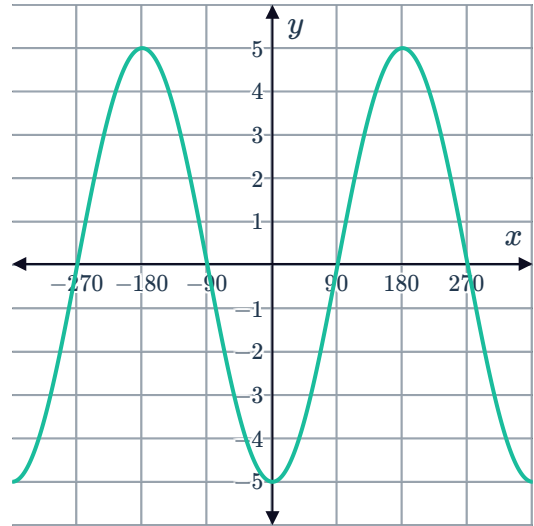


d

i $A = 5$



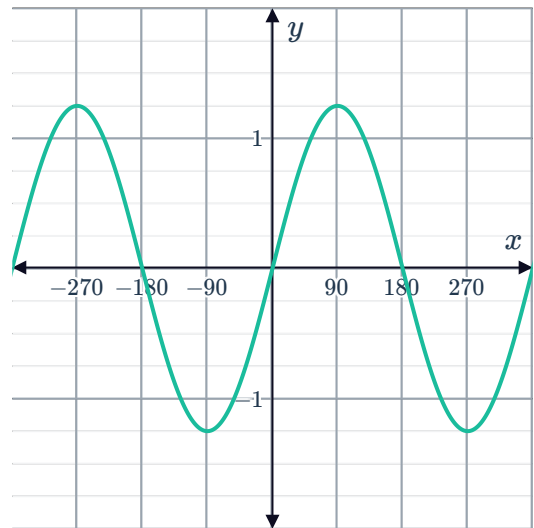
ii



e

i $\frac{5}{4}$

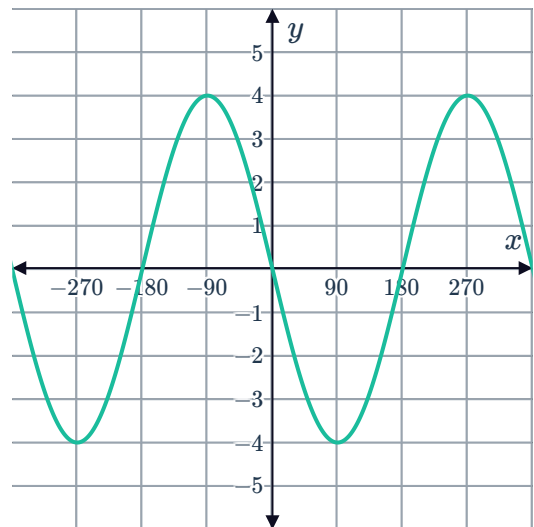
ii



f

i $A = 4$

ii



b -3

c 3

d A vertical dilation by a factor of 3 and reflection about the x -axis.



Vertical translations

12 Translated 2 units downwards

13 Translated 2 units upwards

14 Translated 4 units upwards

15

a $y = \sin x + 4$

b $y = \sin x - 2$

c $y = \sin x - 4$

d $y = \sin x + 1$

16

a $y = \cos x + 3$

b $y = \cos x - 4$

c $y = \cos x - 3$

d $y = \cos x + 2$

17 Reflection about the x -axis, then translation 3 units up.

18 $y = 2$

19

a Translated 4 units upwards

b 360°

c 5

d 3

20

a

i 360

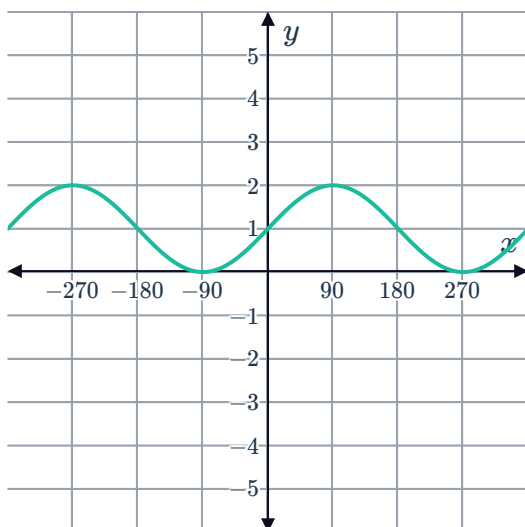
iii 2

v



ii 1

iv 0



b

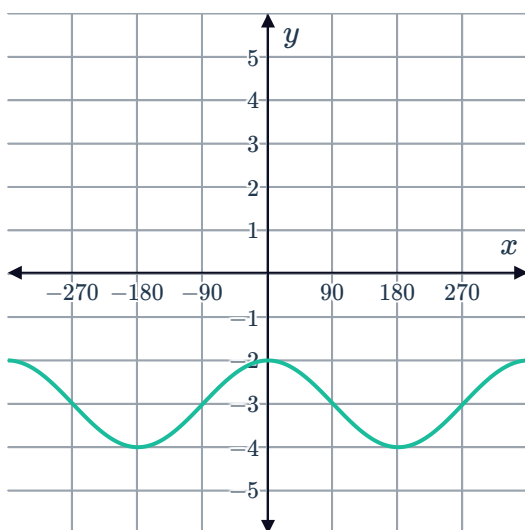
i 360

iii -2

v

ii 1

iv -4

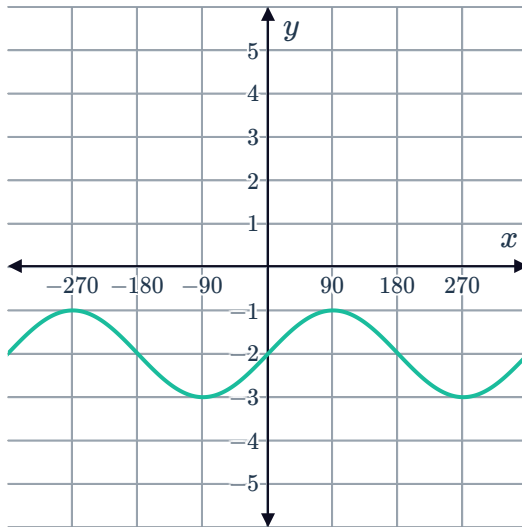


c

i 360

iii -1

v



ii 1

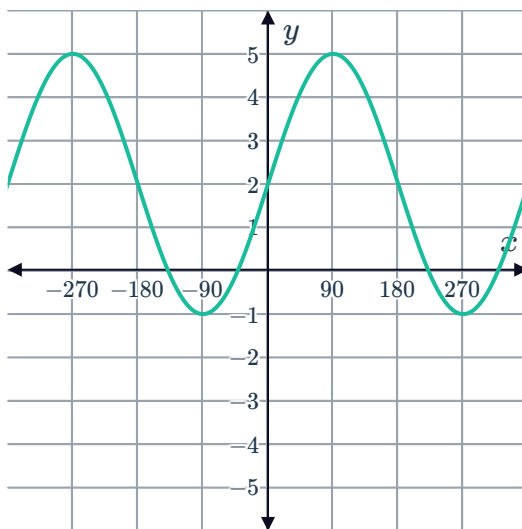
iv -3

d

i 360

iii 5

v



ii 3

iv -1

e

i 360

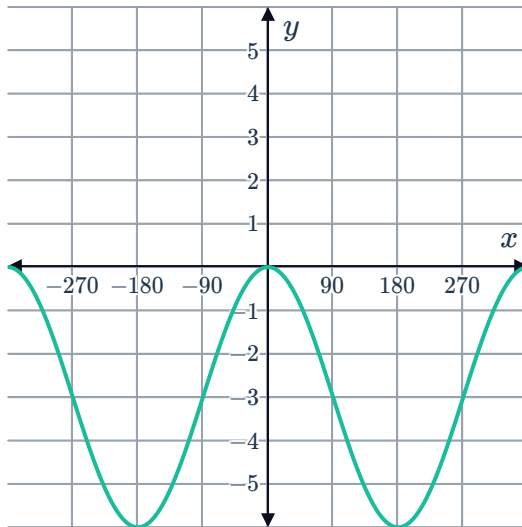
iii 0

v



ii 3

iv -6



f

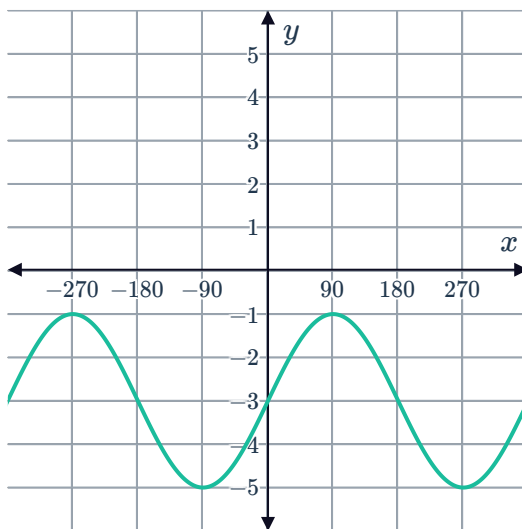
i 360

iii -1

v

ii 2

iv -5



21



a

i 3

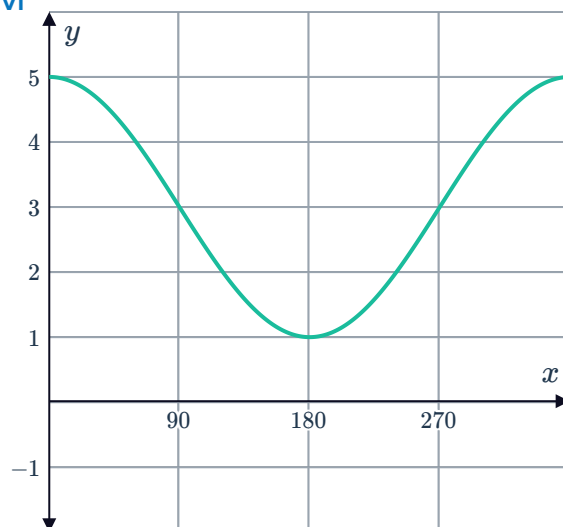
iii 5

v A vertical dilation by a factor of 2, then translated 3 units upwards.

ii 2

iv 1

vi



b

i -3

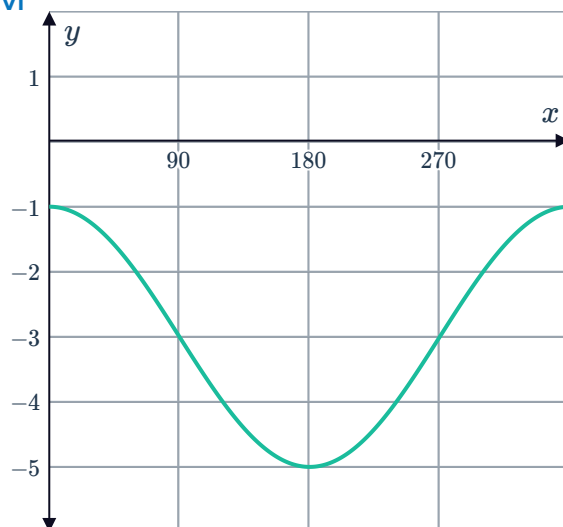
iii -1

v A vertical dilation by a factor of 2, then translated 3 units downwards.

ii 2

iv -5

vi



c

i 2

iii 6

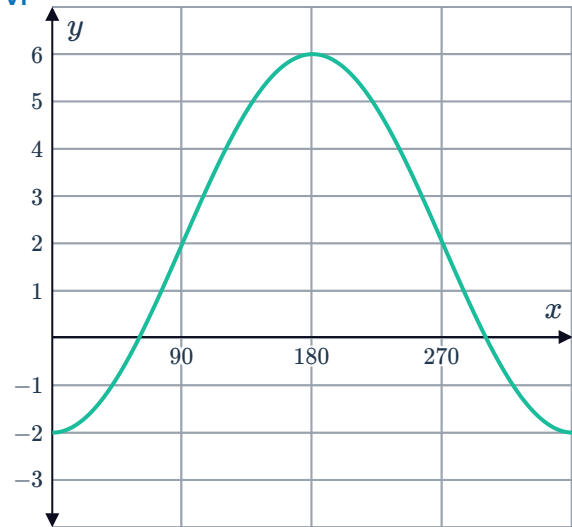
v A vertical dilation by a factor of 4, a reflection about the x -axis and then translated 2 units upwards.



ii 4

iv -2

vi



d

i $\frac{7}{3}$

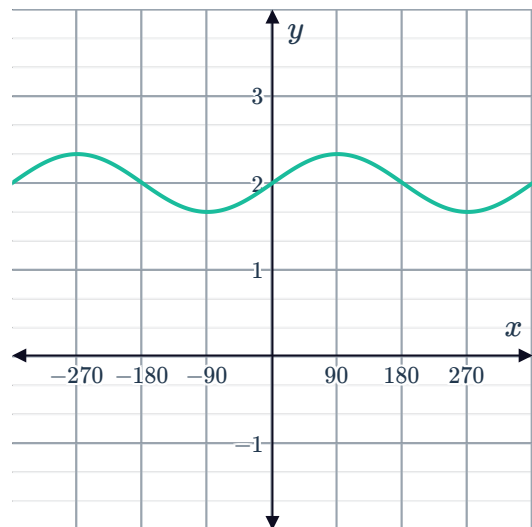
iii $\frac{7}{3}$

v A vertical dilation by a factor of $\frac{1}{3}$, then translated 2 units upwards.

ii $\frac{1}{3}$

iv $\frac{5}{3}$

vi



22

a $y = \cos x + 9$

b 10

23

a

i 5

ii -1

b

i 7

ii 1

c

i 2

ii 1

d

i 2

ii $\frac{1}{2}$

24 $2 \sin x + 2$

25 $8 \cos x - 2$

